

MBS

Modular Building System

version 0.2

made by

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questions? bugs? wishes? send me an email

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About MBS

Modular Building System(MBS) is a level-design tool for Unity Editor.

The purpose of MBS is to quickly place users' prefabs in the scene. Using MBS and modular prefabs (prefabs that are purposely created for multiple repetitions) a user can create 3D buildings or other structures inside Unity Editor, i.e. design game or app level.

Developer: Roman Ponomarenko, Kazakhstan, Ekibastuz.

Minimum Requirements

MBS v0.2 requires Unity 2021. Other versions of Unity may or may not work, but are not supported.

Warning

MBS is in early access and in active development. While using MBS you can encounter errors and incorrect work of MBS or Unity Editor. The developer does everything possible to eliminate such cases.

If you encountered any problem while using MBS, I would be grateful if you inform me about it, see more [Problems Reporting](#).

If you are not satisfied with MBS and want to get a refund, please contact me via email with the purchase information attached.

Glossary

- **MBS** - Modular Building System
- **Modular Prefab** - prefabs that are purposely created for multiple repetitions
- **Asset Pack** – set of Unity assets such as prefab, material, texture, 3d model, etc., and grouped together by some attribute, to use with Unity.
- **Modular Pack** - asset pack that has a descriptor file, which describes to MBS how to use that or that asset of asset pack.
- **Modular Group** - a constituent element of a modular pack that keeps the information about a small group of assets that are grouped by some attribute. More often it is objects that are very similar to each other but perform different functions.

For example, a wall group that contains a few modules: a modular wall with a window, a modular wall with a doorway, and so on.
- **Module** - a constituent element of a modular group that keeps information of one or a few prefabs that represent the same object but with different parameters.

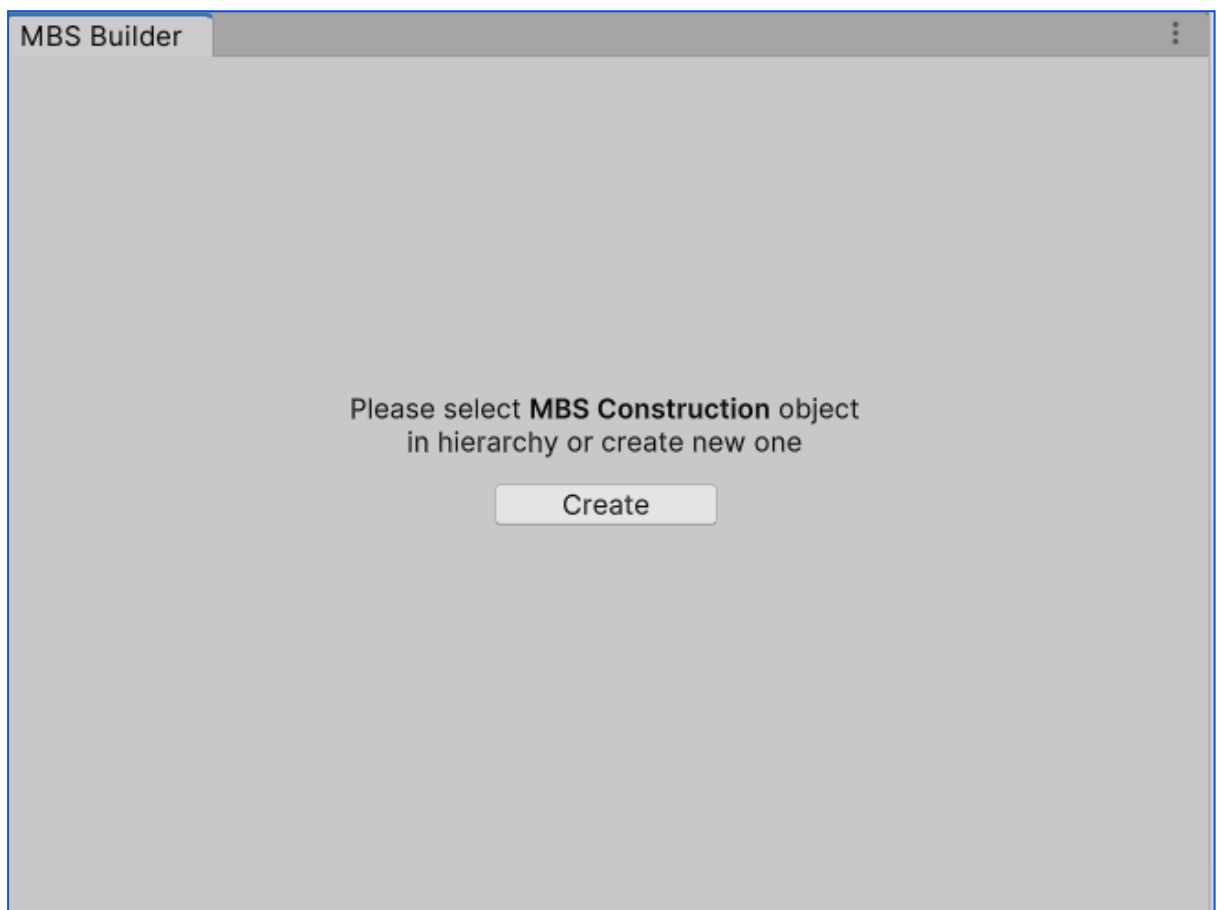
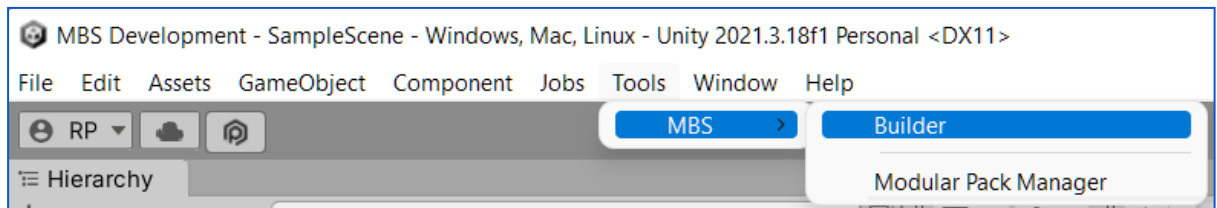
For example, a floor module that contains two prefabs: square floor tile prefab for regular placement and triangular floor tile (the same square floor tile cut in half) for placement in certain situations.

How to Build

Open MBS Builder

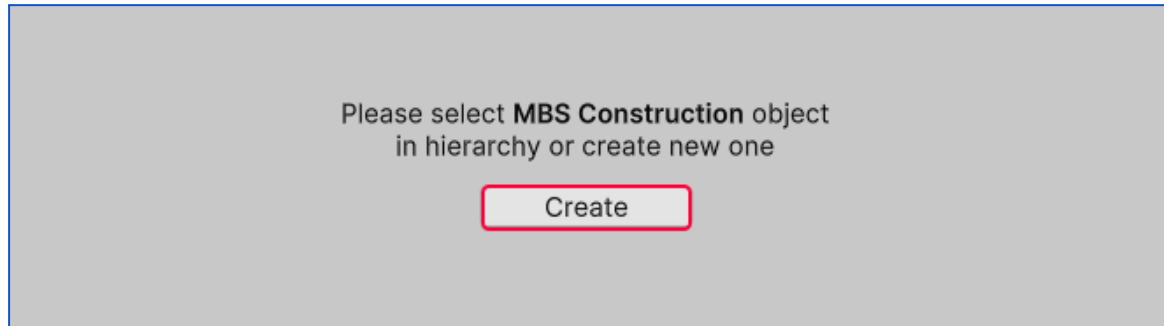
In MBS the construction process takes place through the **MBS Builder** window.

You can open the **MBS Builder** window via the menu: *Tools \ MBS \ Builder*

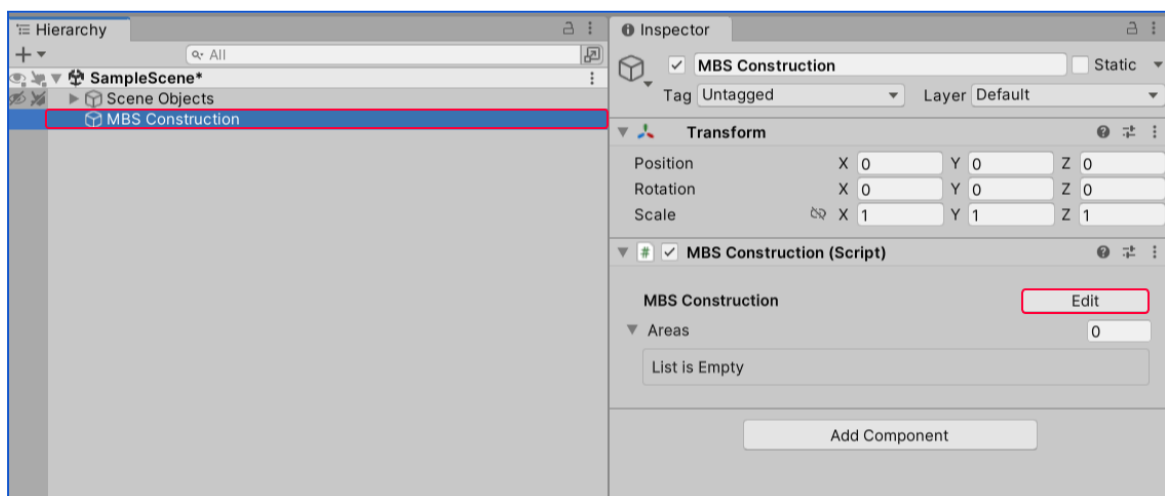


Create MBS Construction

Now, you need to create a new **MBS Construction** game object or edit the existing one. To create a new one, click on **Create** button in the **MBS Builder** window.



* **MBS Construction** is a game object with an attached script which is stores and manages all data that belongs to this particular construction, it is also a container(parent) for all objects related to it.

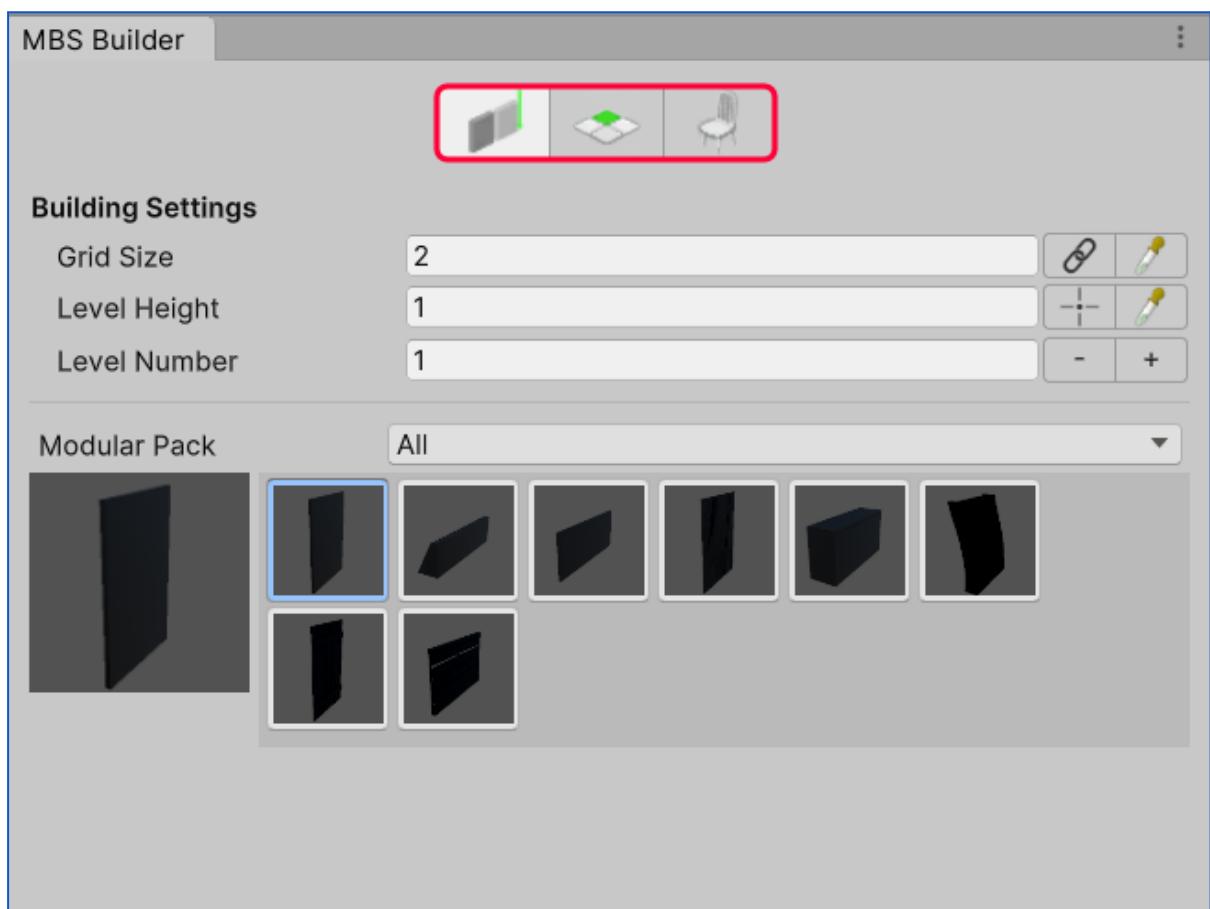


To edit the existing construction, select the game object in the hierarchy and click on the **Edit** button in the inspector window under the MBS Construction component.

Select the tool

When MBS Construction is created and selected for editing you can select the building tool.

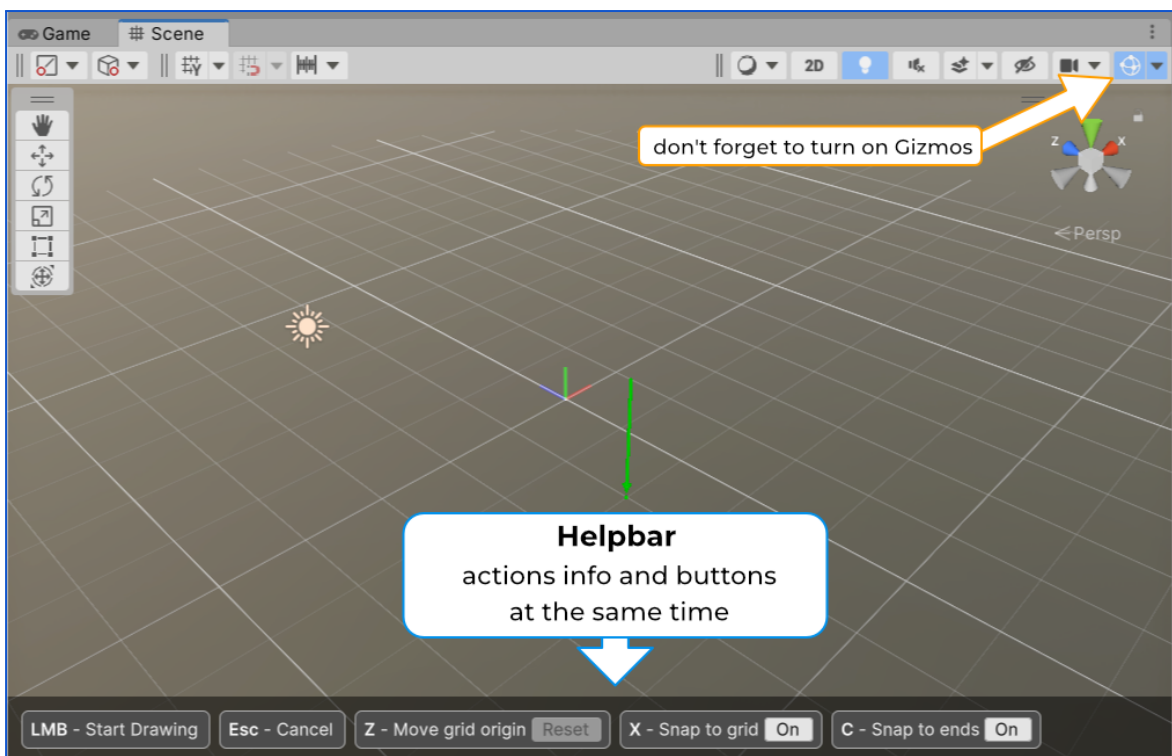
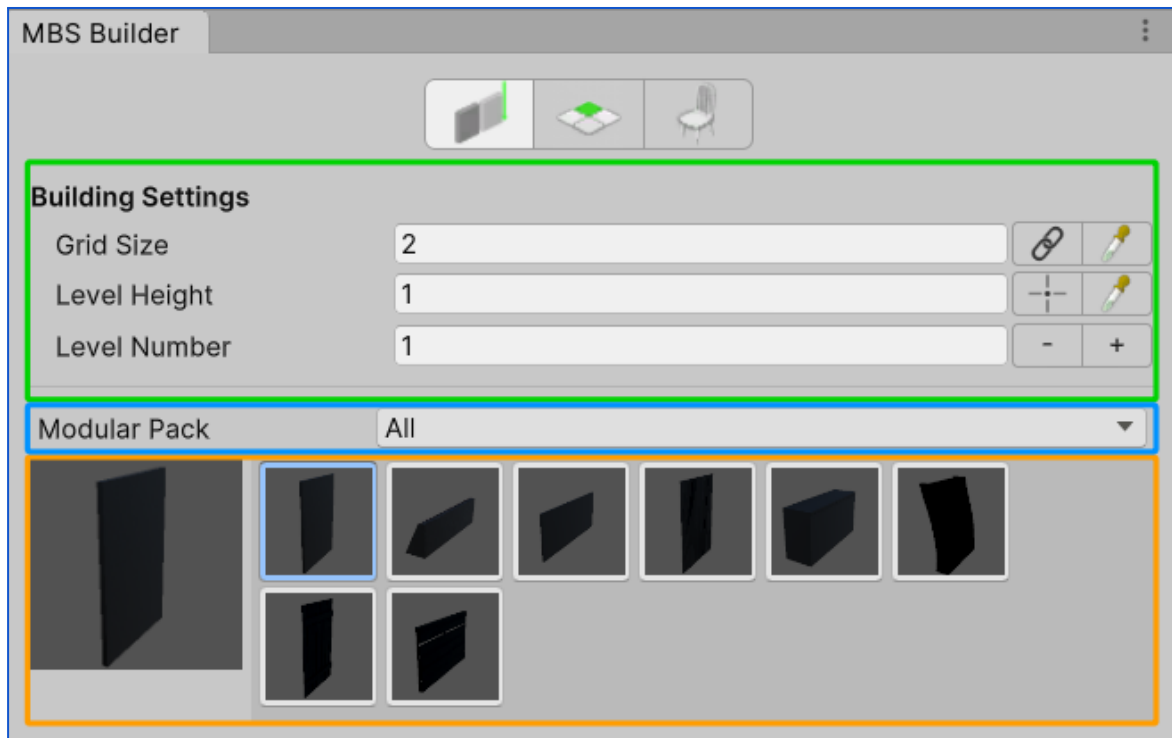
MBS v0.2 contains 3 building tools: **Wall Builder**, **Floor Builder**, and **Decorator**. You can see them in order at the top of the MBS Builder window.



Wall Builder Tool

Building

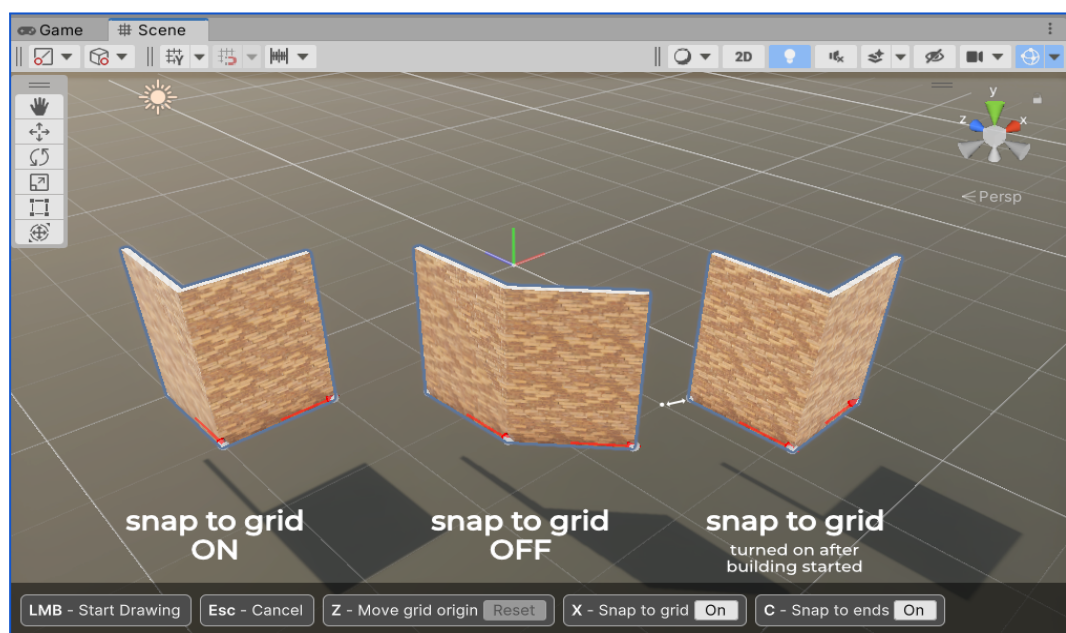
To build a wall, select the needed **Modular Pack**, then select **Modular Group**, then adjust the **Building Settings**, move the pointer into the Scene View window, and start to build.



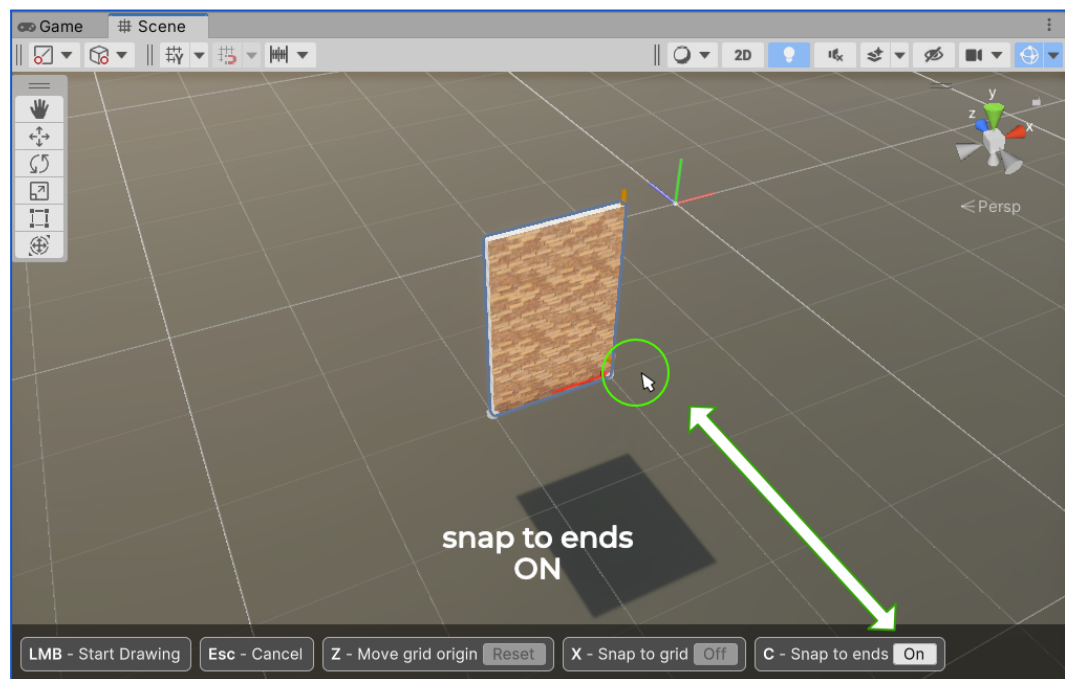
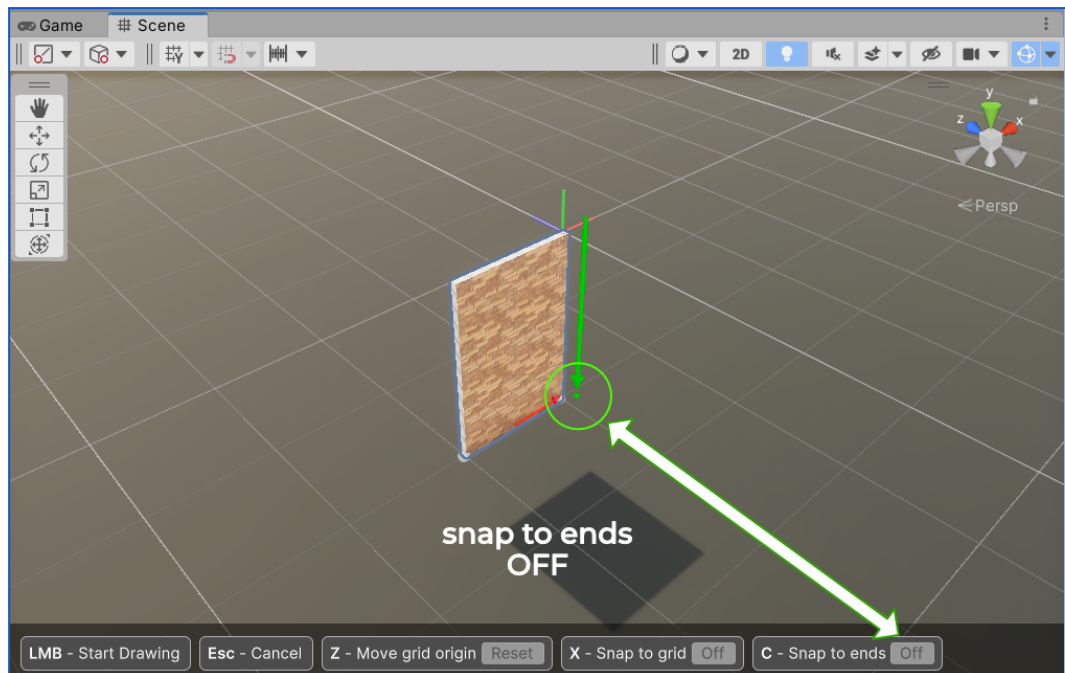
You can see a list of available actions for the currently selected tool in the **helpbar** at the bottom of the Scene view window. Here you can see the **hotkey for each action**, also you can use each element as a **button**, and for some elements, you can see an **additional button or toggle** inside. For example, in the image above you can see that the **Snap to Grid** is enabled and its toggle is **On**. You can click on it and turn it off by mouse, or using correspond hotkey.

Actions:

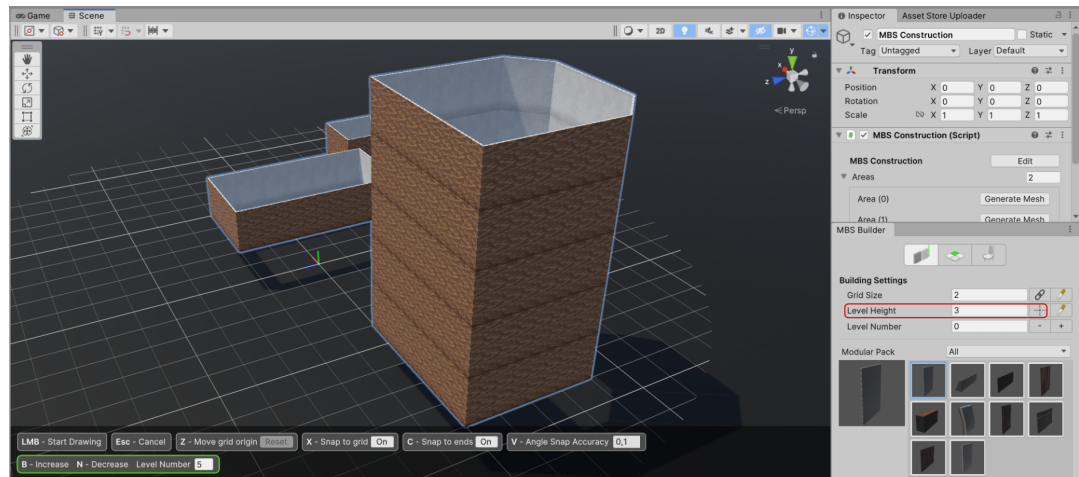
- **Move Grid Origin** - you can move grid origin by moving the pointer in the scene view in the desired place and using a hotkey. Also, you can reset the grid origin using the **Reset** button.
- **Snap to grid** - you can build walls by snapping to the grid or without snapping. You can use this ability to set the pointer in the desired non-grid snapped place, start building, and then turn it on to build straight and snap to the grid wall.



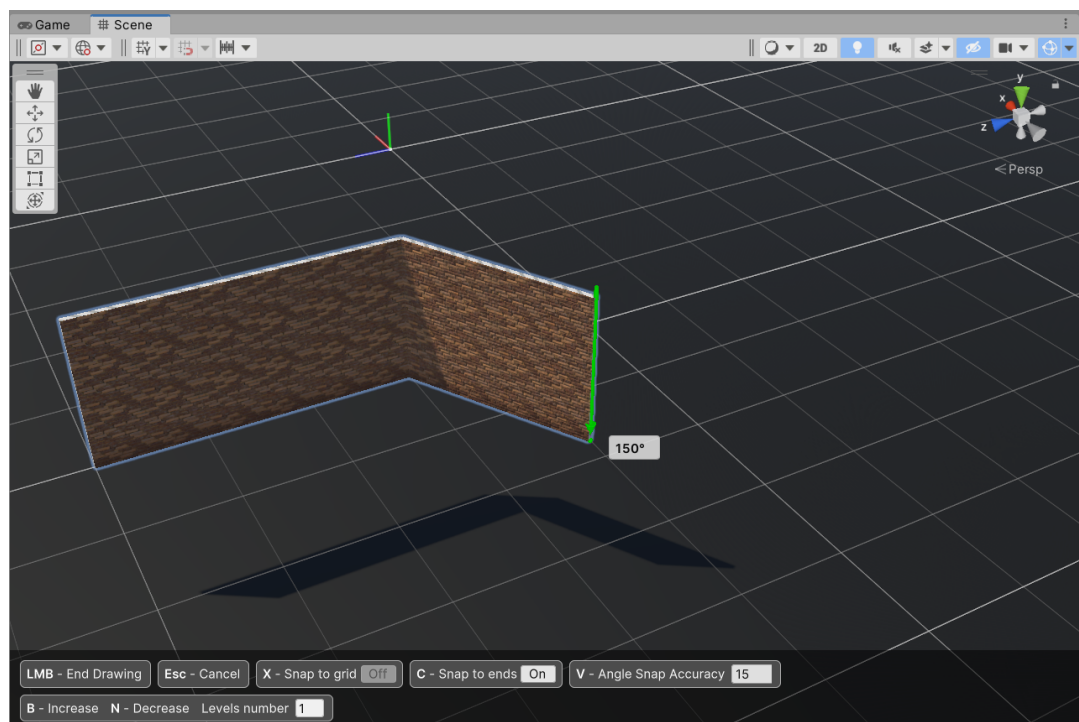
- Snap to ends - snap building pointer to ends of already built walls inside current construction.



- Levels Number - allows you to build multi-level (in other words multi-floor) walls. Firstly you need to adjust the Level Height property in Builder settings (MBS Builder window) and then adjust the level number in helpbar.



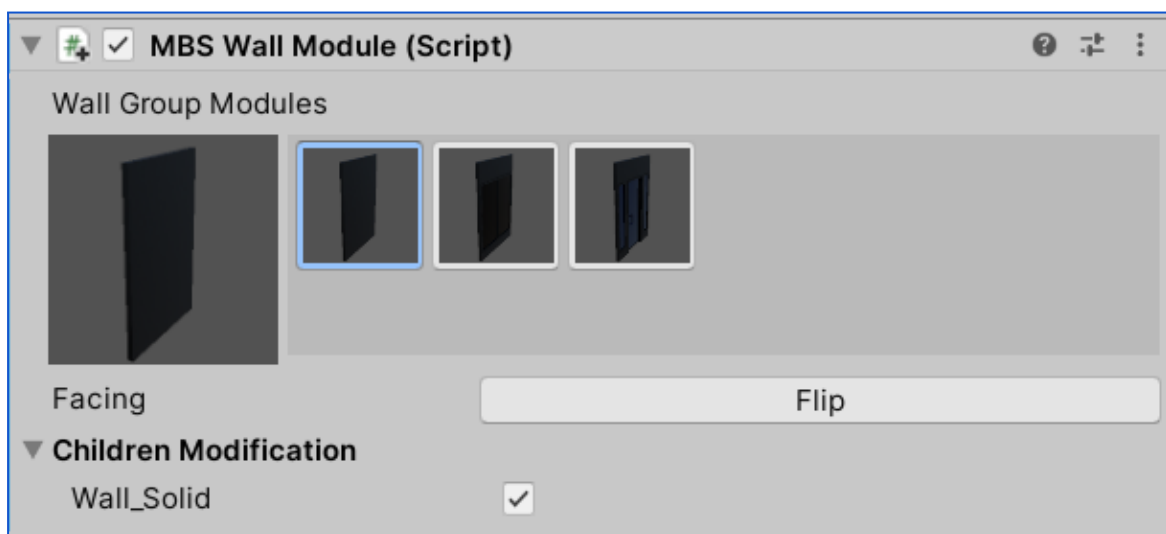
- Angle Snap Accuracy - allows you to snap rotation angle during wall placement. There are 5 options available: 90, 15, 10, 1, and 0.1 degrees.



Editing Wall Module

You can edit placed objects in the scene. To edit the module, select the needed object(s), and proceed with one of the available operations:

- **Change module.** You can change the current module on the other one if the corresponding Modular Group has it.
- **Flip the wall.** If the wall is placed in the wrong way you can flip it.
- **Edit children modification.** During its work, MBS can modify meshes of instances of wall prefabs and their children, if you want to disable this modification for some part of your prefab, you can disable it using the

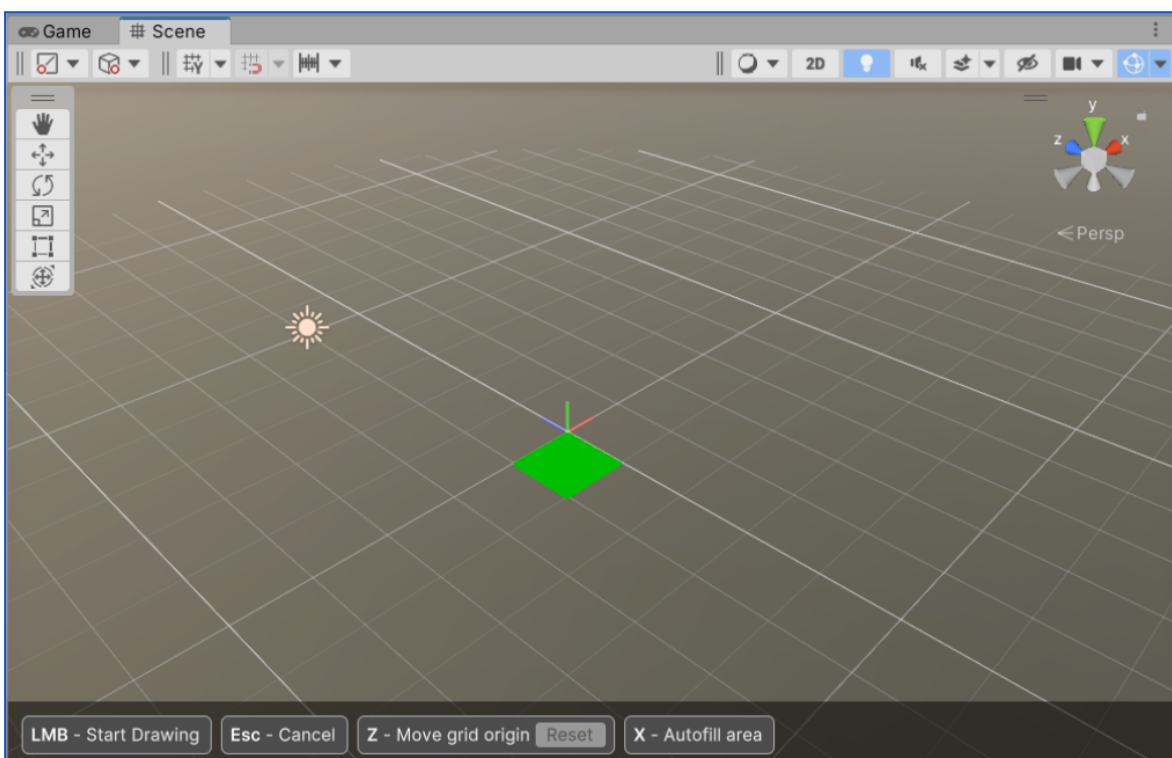
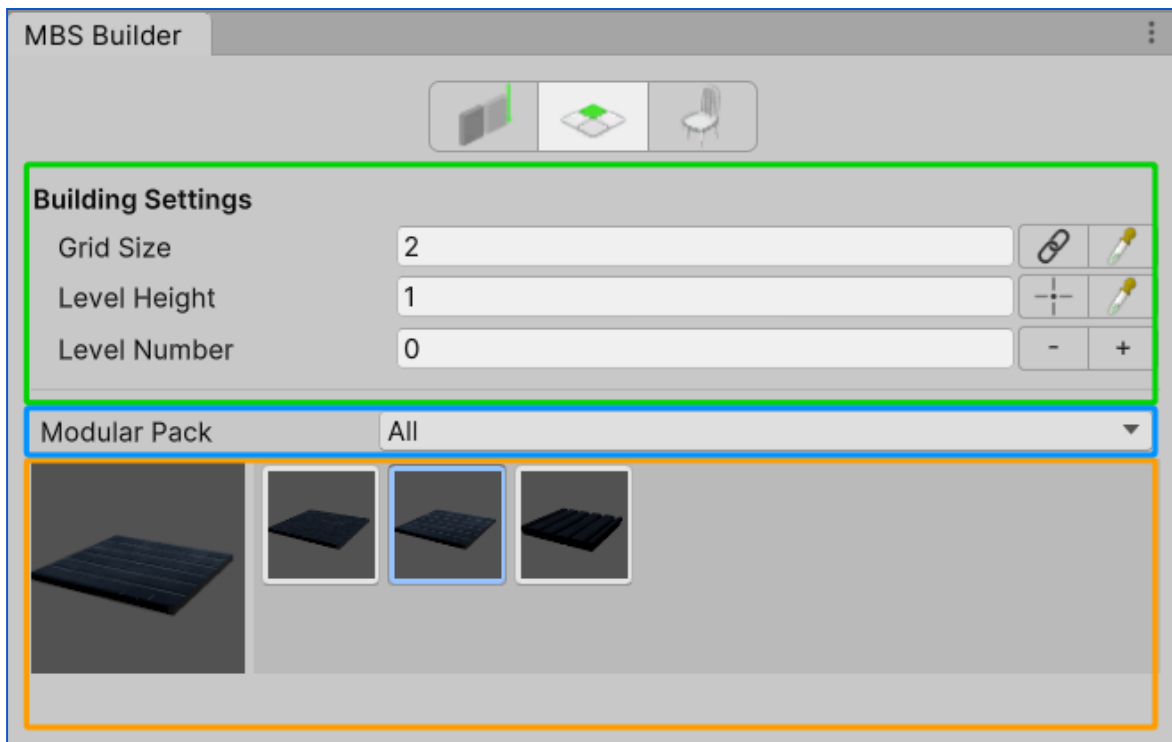


checkbox.

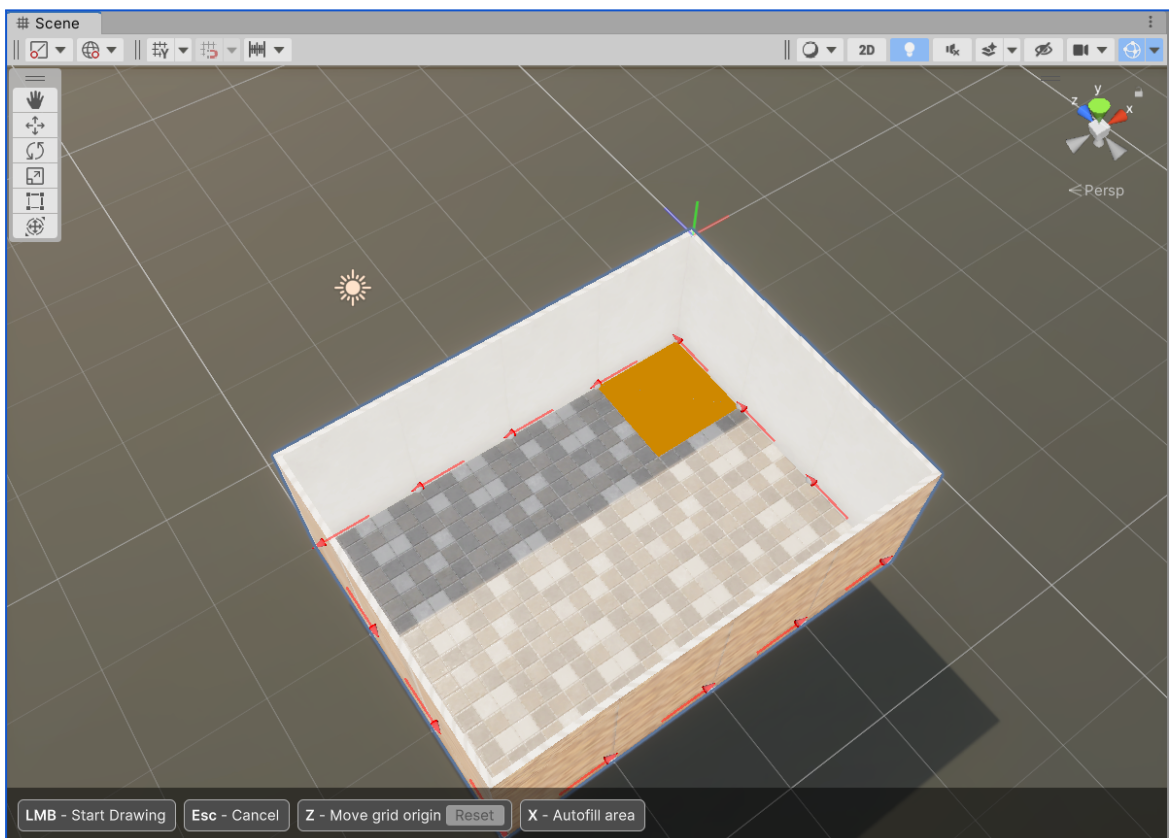
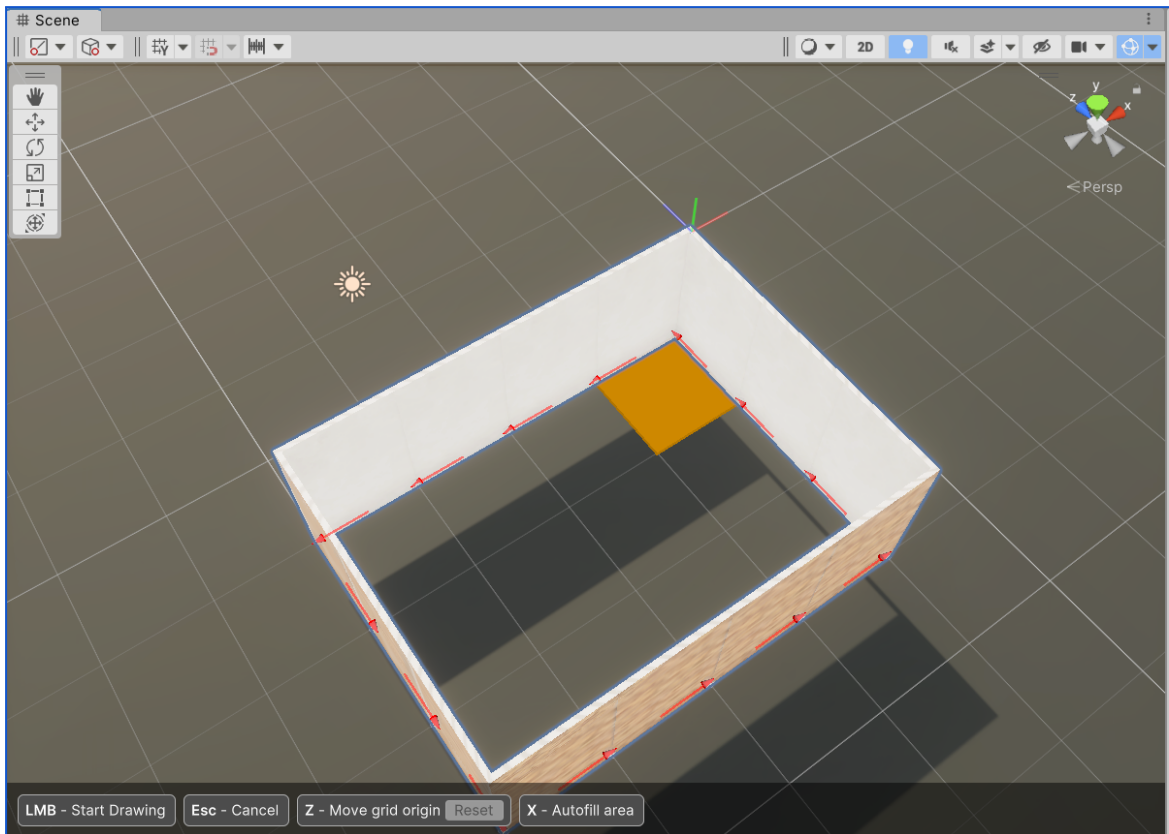
Floor Builder Tool

Building

To build a floor, select the needed [Modular Pack](#), then select [Modular Group](#), then adjust the [Building Settings](#), move the pointer into the Scene View window, and start to build.

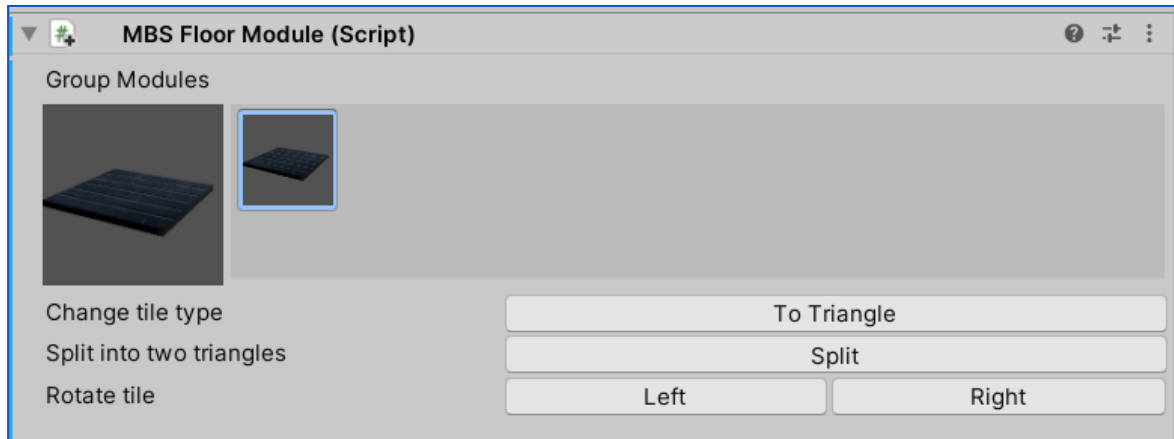


You can see that the Floor Building Tool has **Autofill area** action. Using it you can quickly fill the floor inside the closed area, the only thing you need to know is that you need to use the proper Grid Size value in building settings or stick your pointer to the nearest walls corner, so your floor could be properly aligned.



Floor Builder Tool is heavily dependent on the **Grid Size** parameter, and if the value will be less than the selected module size it will cause overlapping, and one tile will overlap the other.

Editing Floor Module



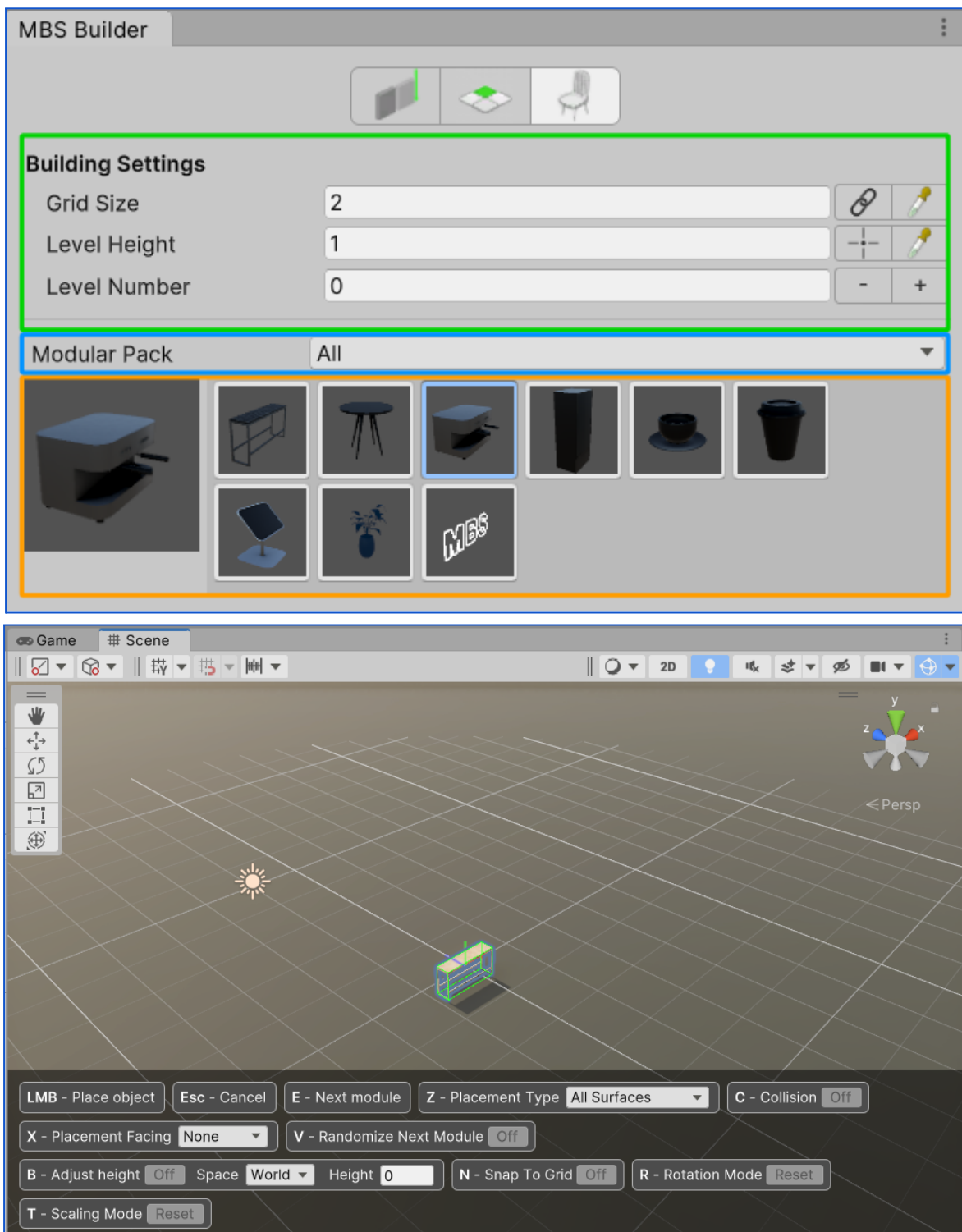
Actions and Parameters:

- **To triangle** - will change the tile type if the corresponding module has it.
- **Split** - when the square tile is selected you can split it into two triangle tiles if the corresponding module has triangle tiles. Can work otherwise.
- **Rotate tile** - will rotate the tile.

Decorator Tool

Building

To decorate, select the needed **Modular Pack**, then select **Modular Group**, then adjust the **Building Settings**, move the pointer into the Scene View window, and decorate.

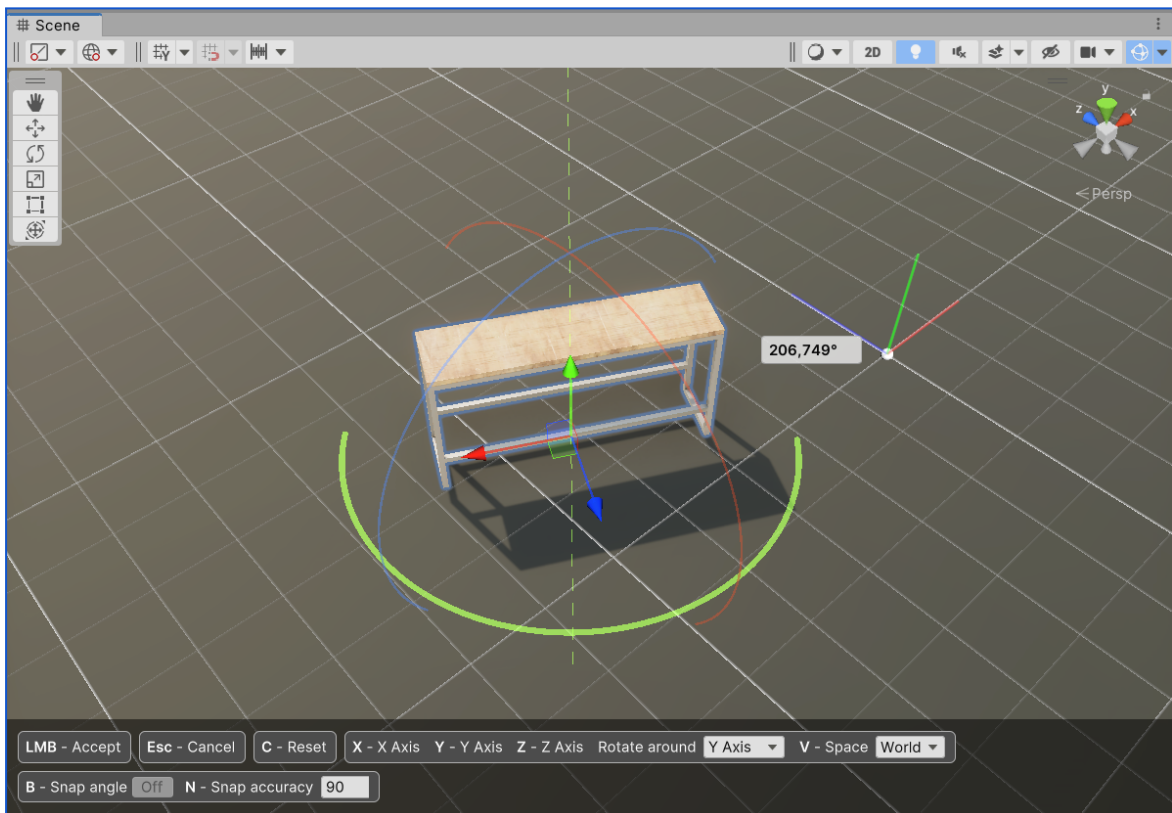


Decorator Tool actions and parameters:

- **Next Module** - will select the next module from the list of the corresponding Decorator Group.
- **Placement Type** - defines the placement method.
 - All surfaces - the object will stick to any surface, whether it be vertical, horizontal, or inclined.
 - Upwards Plus - the object will be projected down to the first obstacle on the way, from any surface that is not upward.
 - Upwards Only - the object will stick only to the upward surface and will ignore all non-upward surfaces.
- **Collision** - enables/disables collision. If the collision is enabled the object will not go through the other objects that have a collider.
- **Placement Facing** - defines which side the object will be facing from the surface.
- **Randomize Next Module** - after placing the current object will select the next random module from the corresponding modular group.
- **Adjust height** - will set a specified height for the object placement, the height is relative to the selected space. For World Space - from world 0, for Self Space - from the current Level height.
- **Snap to grid** - will snap object placement to the nearest grid cell position.
- **Rotation Mode** - will turn on the rotation mode.
- **Scaling Mode** - will turn on the scaling mode.

Rotation Mode

This mode will allow you to rotate your object before the placement.

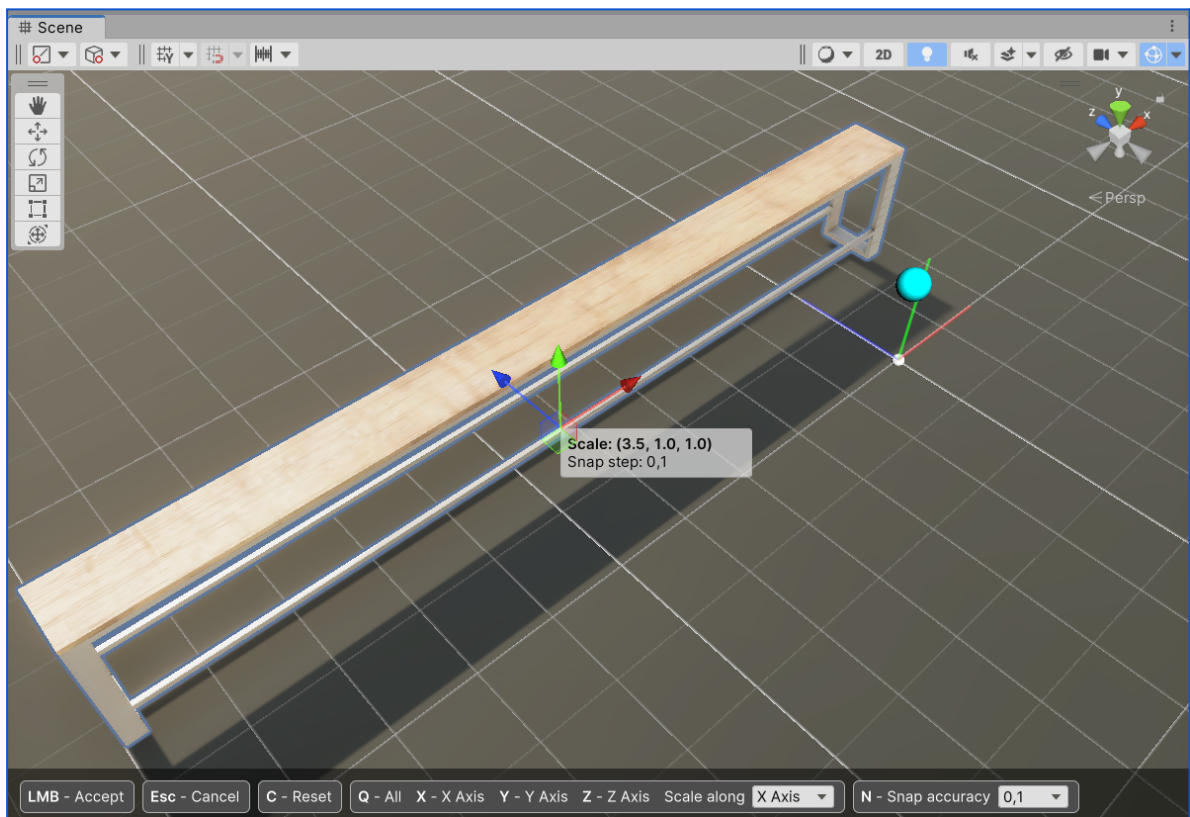


Rotation Mode actions and parameters:

- **Reset** - will reset the rotation of the object to the initial state.
- **Rotate Around** - defines the axis around which rotation will proceed.
- **Space** - defines the rotation space.
 - World - rotation around the world X, Y, and Z axis.
 - Self - rotation around the local X, Y, and axis.
- **Snap angle** - enables/disables snapping rotation angle.
- **Snap accuracy** - defines the snap step. 4 options are available: 90, 15, 1, 0.1

Rotation Mode

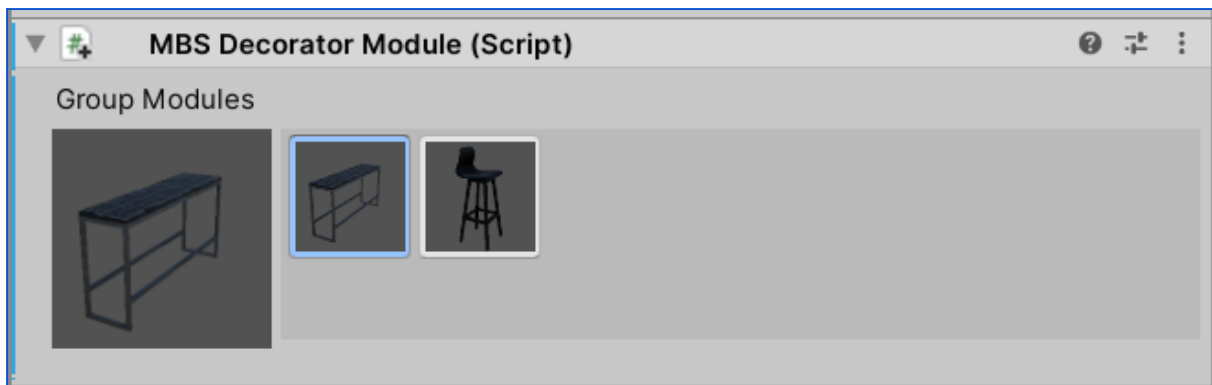
This mode will allow you to scale your object before the placement.



Scaling Mode actions and parameters:

- **Reset** - will reset the rotation of the object to the initial state.
- **Scale Along** - defines the axis along which scaling will proceed.
- **Snap accuracy** - defines the snap step. 3 options are available: 0.1, 0.01, 0.001

Editing Decorator Module



In the Decorator Module only one action is available - change the module.

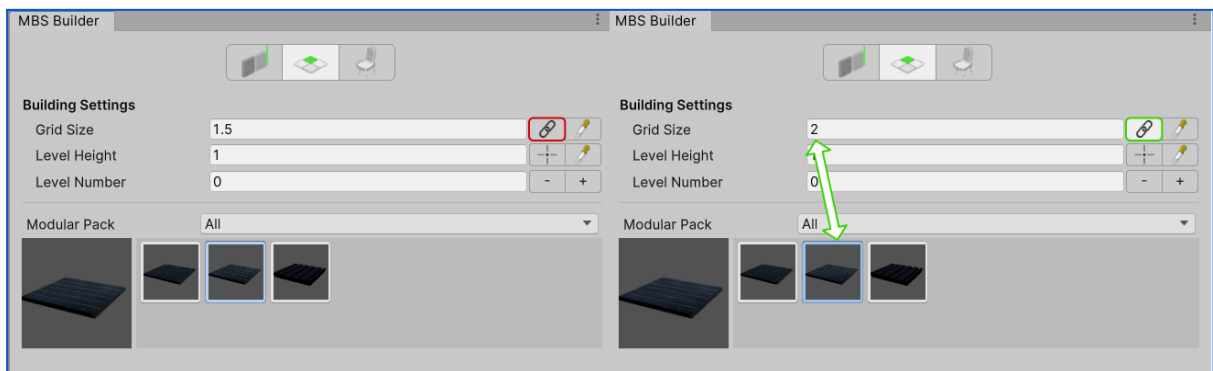
Building Settings

Building Settings			
Grid Size	2		
Level Height	1		
Level Number	0		

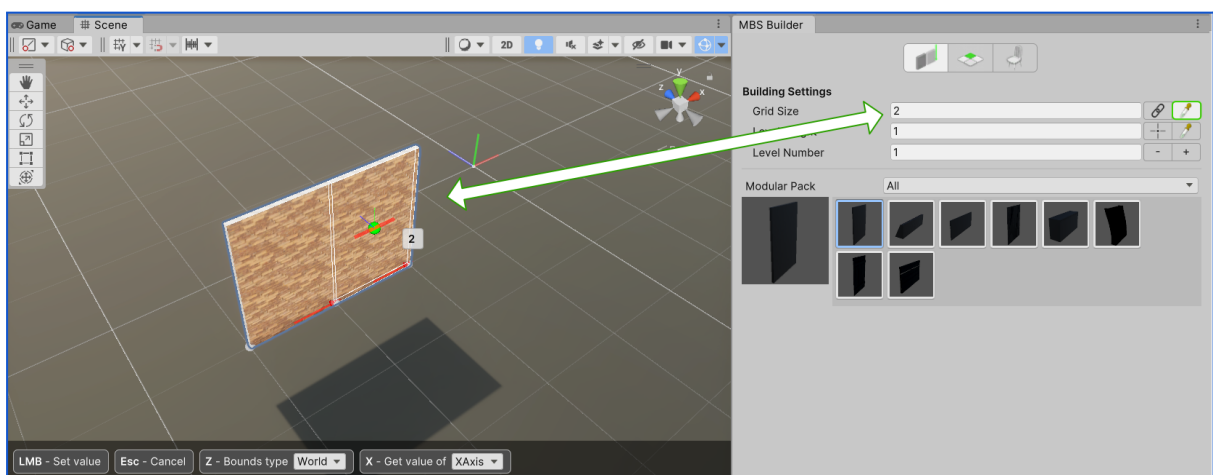
For each building tool, MBS keeps its own Building Settings values. It means that if you set your parameters in Wall Builder Tool they exist only for this tool, other tools have their own independent sets of values.

Grid Size

- Grid Size field - changes the grid cell size. Which is affect the snap step size, and propagation step (in the wall and floor builder tools).
- Link size button - gets the x-axis size of the selected asset and sets it into the Grid Size field.



- Object picker button - gets the size of the object selected in the scene and sets it into the Grid Size field.



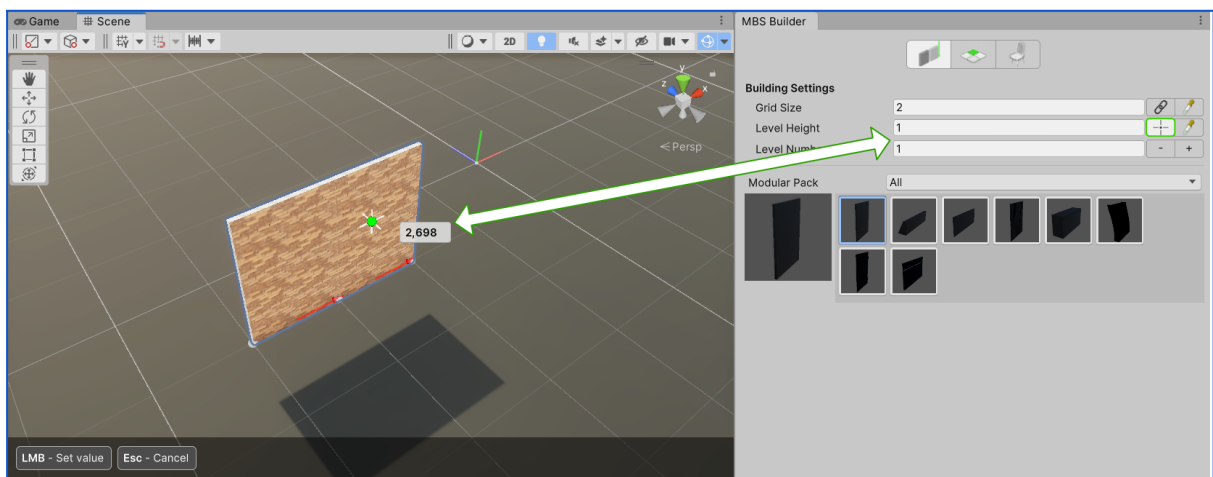
Level Height and Level Number

To work with height levels (floors, stories) MBS uses two parameters **Level Height** and **Level Number**.

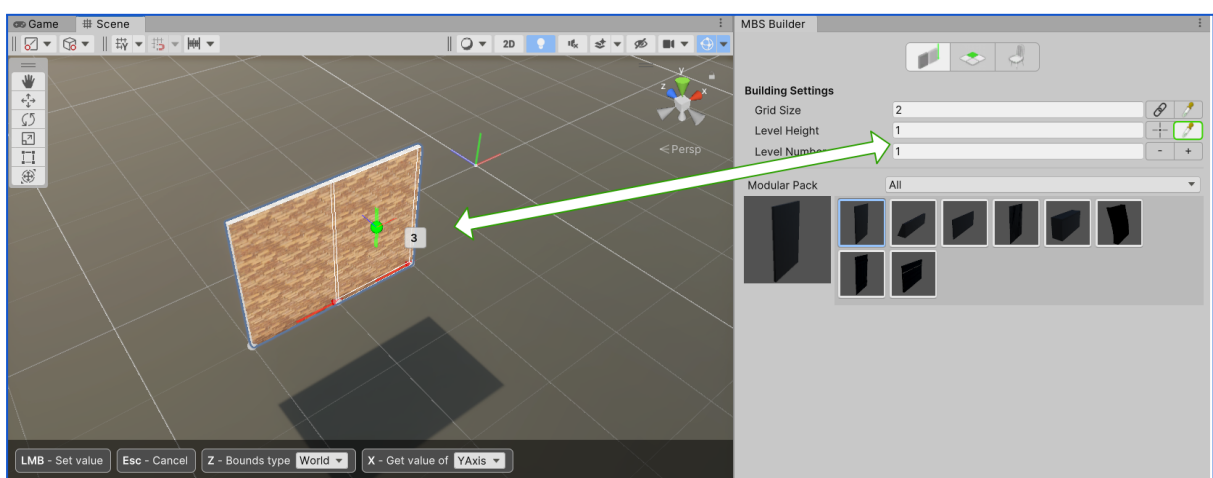
For example, if you are making a building with 3 floors and each one has a height of 3 meters, then you need to set the Level Height to 3, and during the building process increase the Level Number one by one up to 2 (0 is the first level).

So, the **Final Level Height = Level Height * Level Number**

- **Pointer button** - set the level height value from the pointer height in the Scene view.



- **Object picker button** - gets the size of the object selected in the scene and sets it into the Grid Size field.

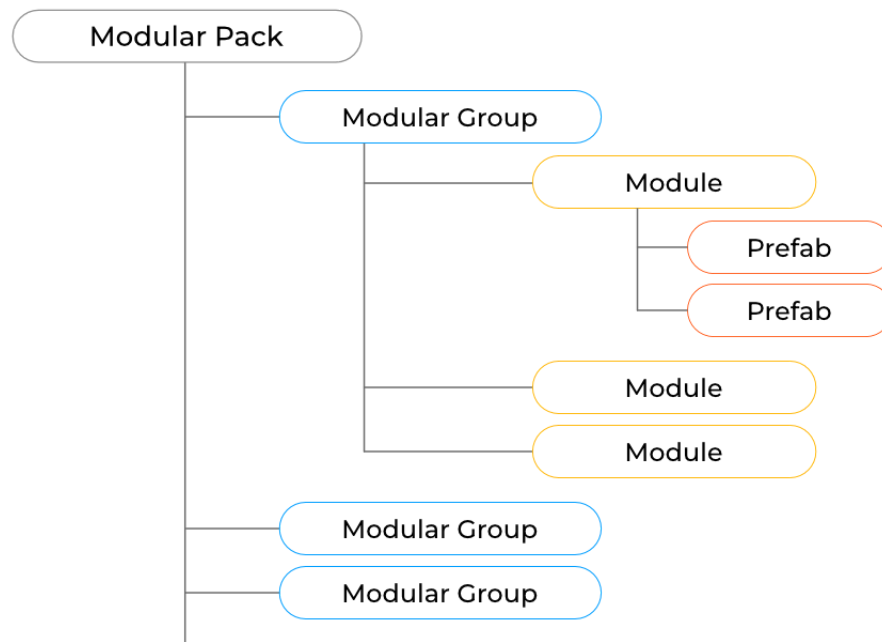


Assets Importing

Data Structure

The main feature of MBS is that it can use custom assets to create buildings and structures.

To work with custom assets MBS has its own data structure which is important to understand.



Each **Modular Pack** contains **Modular Groups**, each group includes **Modules**, and each module consists of one or few **prefabs**.

This may look confusing, but later you will see that it is pretty simple and pretty logical.

Data Structure Understanding

You may have struggled to understand with understanding MBS data structure. It is easy to understand the **Modular Pack** from one side and the **Prefab** from the other, so Modular Pack contains prefabs, but why are there **Modular Groups** and **Modules** in the middle?

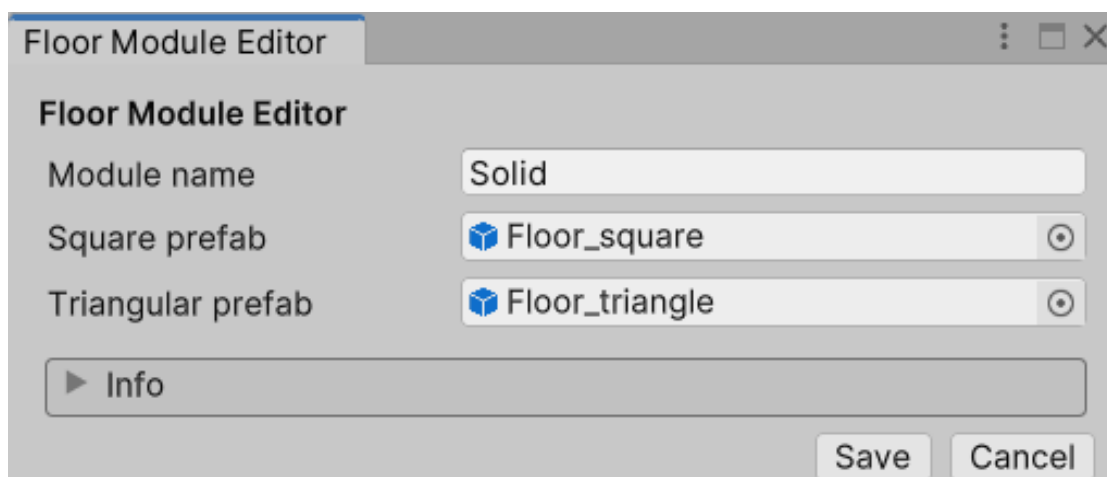
Modular Groups are needed to combine similar elements and elements that can be grouped according to some attribute.

For example, let's say we have 3 similar prefabs of some fantasy stonish wall: a solid one, one with a window, and one with a door. Then, let's say we have a few more wall types, wooden, bricks, sky-fi, and so on. If each wall type will consist of 3 prefabs it will be at least 12+ separate prefabs. Finding a needed wall in such a list could be pretty challenging. This is where **Modular Groups** come to the rescue.

Now, you can create one group for one wall type, one more group for another, and so on.

The **Module** is the minimum unit of the MBS data structure and may contain one or more prefabs that represent a given module but may have different geometry or other parameters.

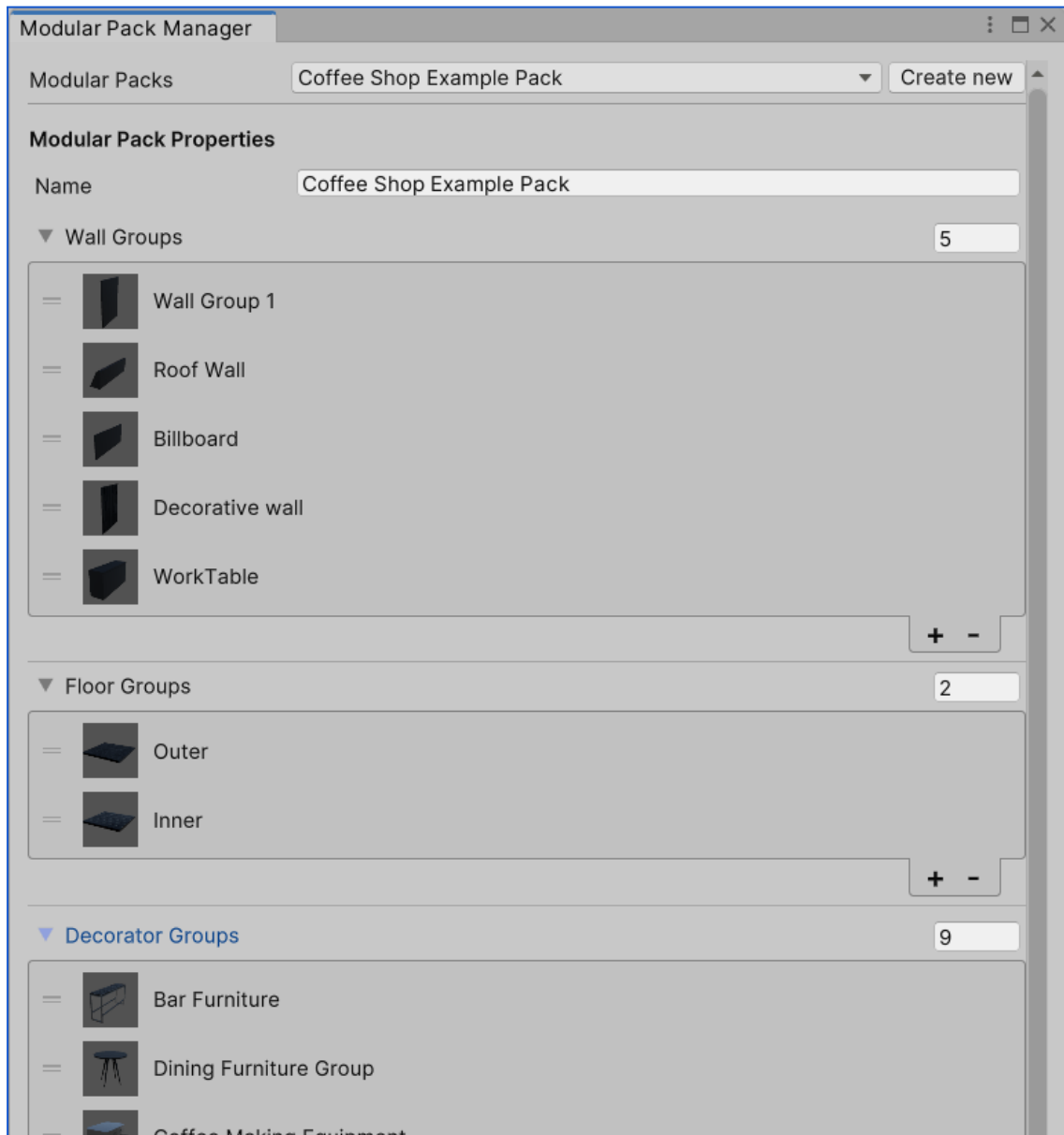
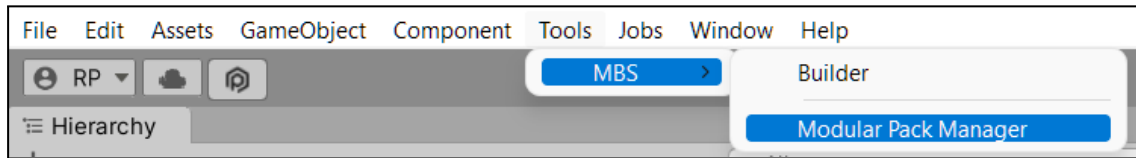
A good example is the floor module, which contains 2 prefabs, one square and the other triangular, but both represent the same module but in different conditions.



How to import assets

1. Open the **Modular Pack Manager** window.

[Tools / MBS / Modular Pack Manager](#)



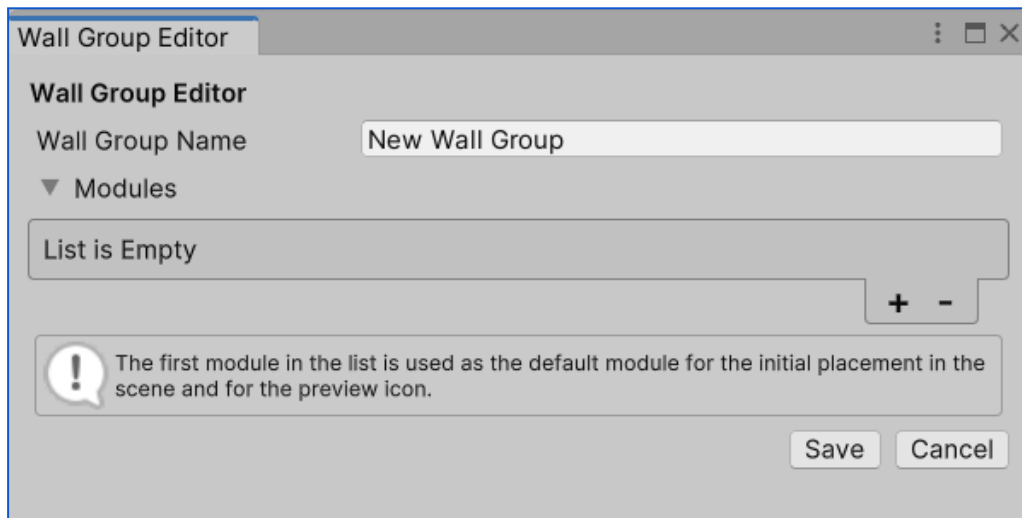
2. Select the Modular Pack you want to edit or create a new one using the corresponding button.

The screenshot shows the 'Modular Pack Manager' window. At the top, there's a tab labeled 'Modular Pack Manager'. Below it, a dropdown menu shows '[Unsaved] New Modular Pack' with a 'Create new' button to its right. The main section is titled 'Modular Pack Properties'. It has a 'Name' field with 'New Modular Pack'. Below this are three sections: 'Wall Groups' with a count of 0 and a 'List is Empty' message; 'Floor Groups' with a count of 0 and a 'List is Empty' message; and 'Decorator Groups' with a count of 0 and a 'List is Empty' message. Each section has a '+ -' button. At the bottom, there are buttons for 'Remove pack', 'Hide', 'Save', 'Discard', and 'Close'.

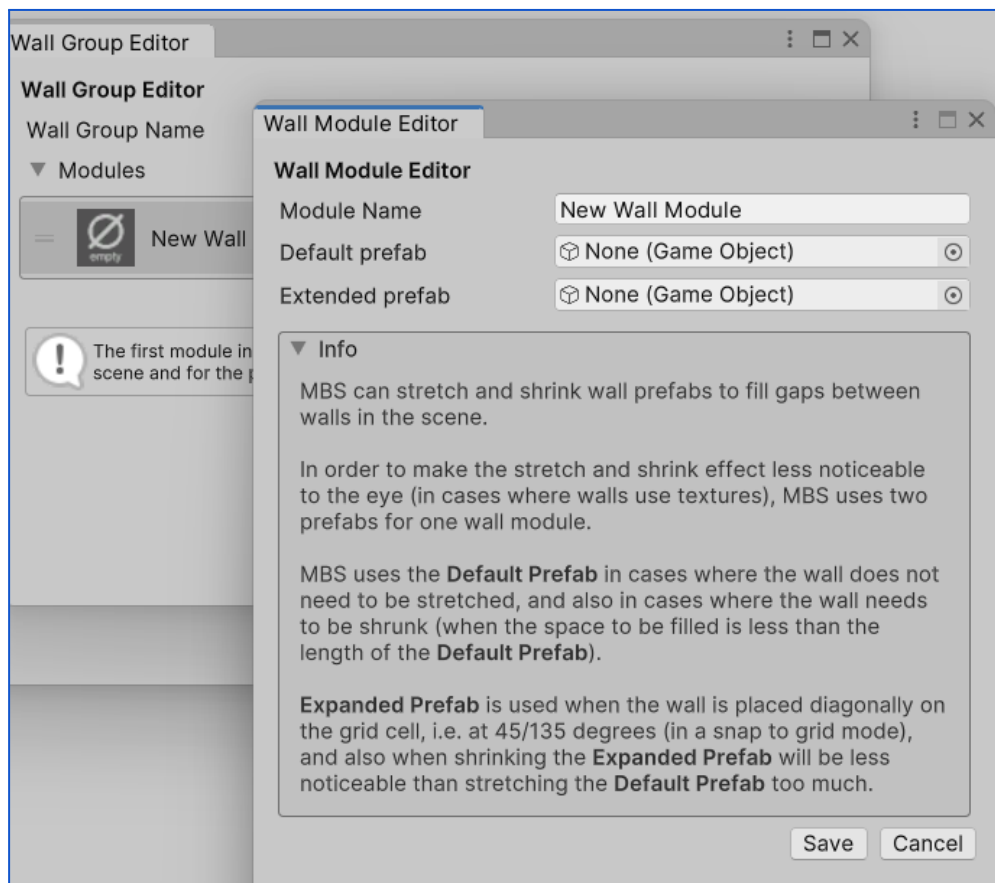
3. Add a Modular Group for the needed tool and double-click on it to edit.

The screenshot shows the 'Modular Pack Manager' window after adding a new wall group. The 'Wall Groups' count is now 1. A new entry, 'New Wall Group', is listed with an 'empty' icon. The 'Floor Groups' and 'Decorator Groups' sections remain empty. The bottom buttons are the same as in the previous screenshot.

4. In opened Wall Group Editor window add a new module and double-click on it to edit.



5. Set the default prefab and click the **Save** button in all opened windows: Module Editor, Group Editor, and Modular Pack Manager.



6. That's all, you've just imported your prefab to MBS. Repeat the same procedure for all needed tools.

Assets preparation

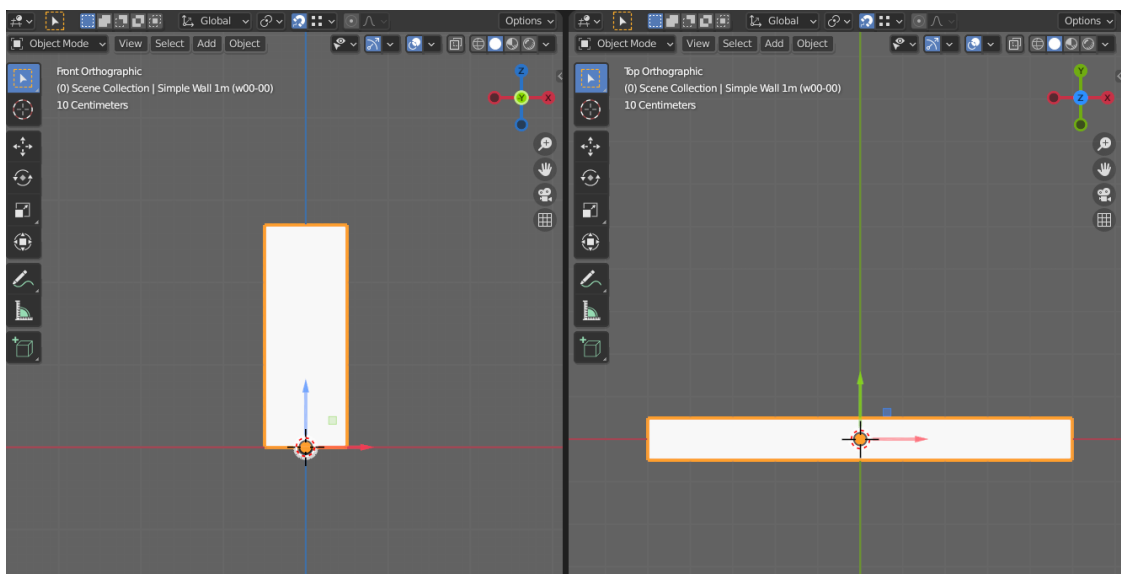
MBS can work with any prefab, but in order to work properly and use all the functions of MBS, you may need to make some changes to the prefab.

Wall

Pivot Point (Origin)

The pivot point is the most important thing. **If you can** work in 3d software like Blender, Maya, and similar you can change pivot points there, **but if you can't** then you can mimic the pivot point in Unity using an empty game object as a parent for your prefab.

For walls - place the pivot in the center bottom of the wall.



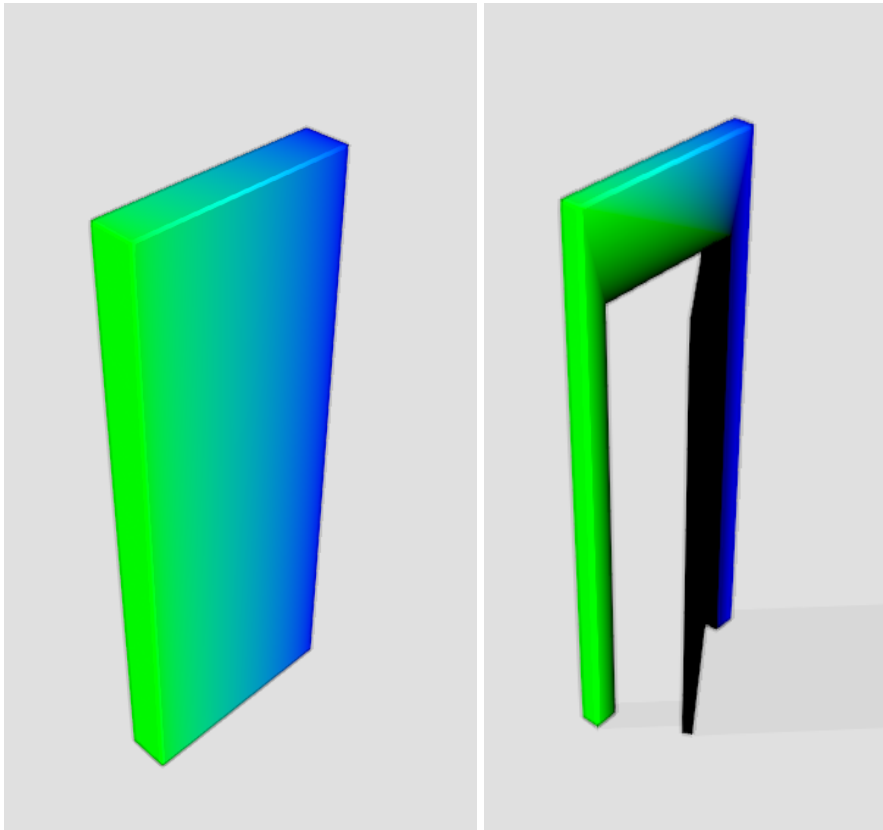
Vertex color

MBS can shrink or stretch wall ends to connect them to hide the gap that occurs at the point of connection. To understand what vertices to touch MBS uses vertex colors.

So, for both ends of your wall, you need to paint the vertices in two different colors, one color for one end, and one for another. For example - the front end will

be $\text{RGB}(255, 0, 0)$, and the other end will be $\text{RGB}(0,0,255)$. The key thing is that one end should have only one color, and each vertex of that end should have this color, even small differences like $(0,0,255)$ and $(0,0,254)$ will be treated like different colors, i.e. different sides.

Also, you can mark vertices that you don't want to be touched, for example, a doorway, or a window, just paint corresponding vertices in black $\text{RGB}(0,0,0)$.

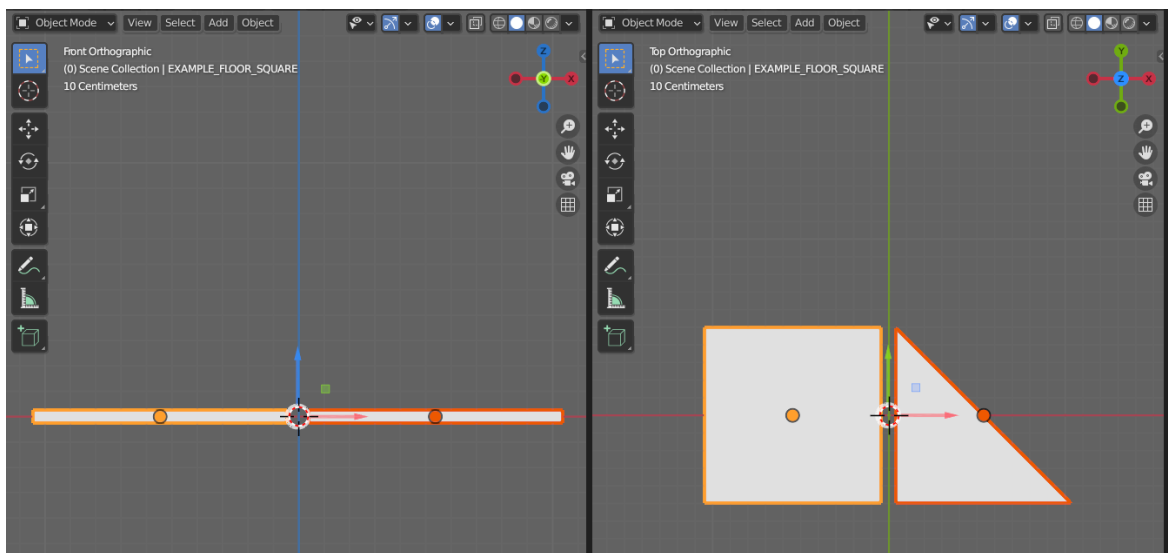


* In the image above you can see a simple 3D model with painted vertices. You can see that in this particular case, the whole model looks painted, but this is only because the model has a small number of vertices. In the case of more complicated 3d models, some parts of the model will not be painted i.e. will have white color.

Floor

Pivot Point (Origin)

For floor tiles make sure your pivot point is in center horizontally. But you can variate it vertically to achieve special results. For example, if your floor tile has a thickness of 0.2m, then if the pivot point is in the middle vertically the floor will have 0.1m above the ground level and 0.1m below. If the pivot point is at the bottom vertically, then the floor will have 0.2m above the level and nothing below, and otherwise, if the pivot point is at the top vertically.



Decorator

Pivot Point (Origin)

The Decorator Tool is the least demanding in terms of pivot points. But even so, it is better to place the pivot point closer to the point of contact with another surface. For example, for a chair, it is better to place the pivot point in the bottom center, for a painting to the center but on the back, where it will be attached to the wall.

Input Configuration

One of the main objectives of MBS is a **fast** level design. To achieve that MBS is focusing mostly on physical inputs i.e. keyboard and mouse rather than UI Buttons.

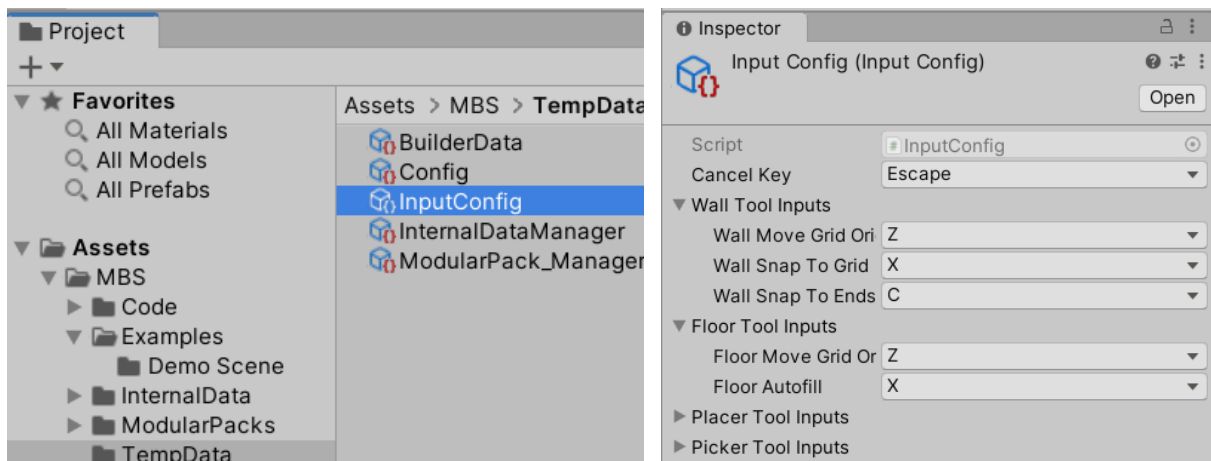
Each tool of MBS (Wall, Floor, and Decorator) has its own set of actions, parameters, and hotkeys for them. By default, all hotkeys are set to the bottom row of the keyboard i.e. Z X C V B N M.

If some hotkeys are not comfortable for you, or they cause some conflicts with other hotkeys, then you can change them.

You can find the **InputConfig** file by this path:

[*MBS / TempData / Input Config*](#)

select this file in Unity Editor, in the Inspector window find the needed action for the needed tool and change it to the key you need.



Problems reporting

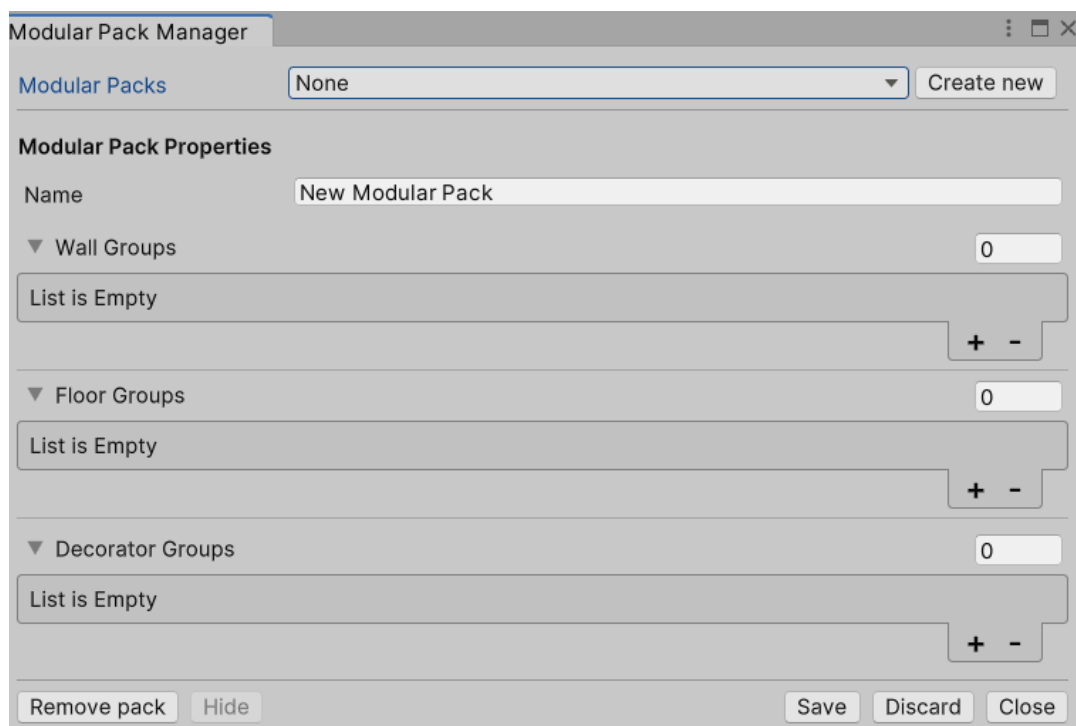
If you encountered some errors or incorrect work of **MBS** or **Unity Editor with MBS**, you can report a problem via email so I could fix it, or give you a hotfix solution.

How to report a problem?

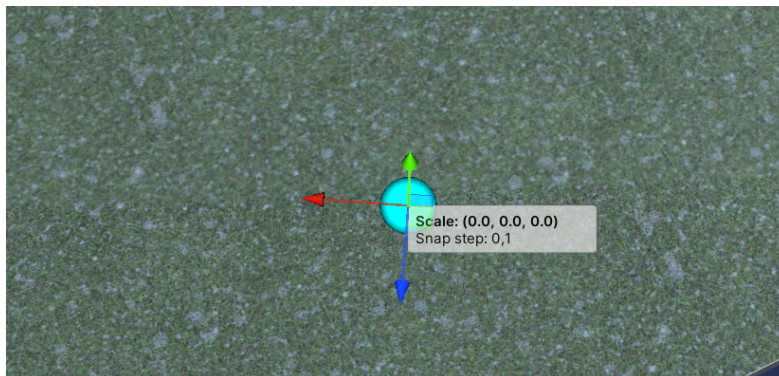
1. First of all, try to understand under what circumstances the problem occurs, i.e. how to reproduce the problem.
2. Make sure that problem is related to MBS.
 - a. Create a new **clean Project** in the **compatible version of Unity**.
 - b. Import **MBS** using **Unity Package Manager**
 - c. Try to reproduce the error or incorrect behavior
 - d. If you **can't** reproduce the problem, then the error may be caused by other assets, plugins, or scripts. **If nothing helps** you, and the problem still occurs while using MBS in your work project, then let me know.
 - e. If you **can** reproduce the problem, follow next.
3. Now that you are sure that the problem is related to MBS and you know how to reproduce it, send me an email and **attach at least something from the list below**.
 - a. **One or a few screenshots** that can clearly show an error message, incorrect behavior, or the problem, and/or how to reproduce the problem.
 - b. **Short screen recordings** that can clearly show an error message, incorrect behavior, or the problem, and/or how to reproduce the problem. For video recording I recommend OBS, it is a free software.
4. Attach the following information:
 - a. Your operational system: Windows, MacOS, Linux
 - b. Your Unity version

Known Issues

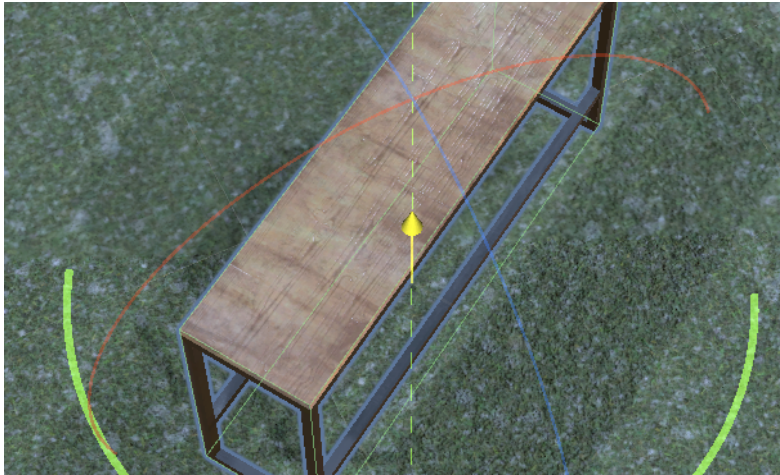
1. **Modular Pack Manager Window:** List View Items are sticking to the mouse pointer and do not go back in place. [Looks like this](#). This one is **Unity Bug**, here is the [IssueTracker Page](#), **please vote for it so they could fix it faster**.
2. **Modular Pack Manager Window:** if click on the “Create New” button, and without saving the new modular pack click on “Create New” once again it will break the window logic.



3. **Decorator Tool:** Scaling Mode is starting with prefab scaled down.



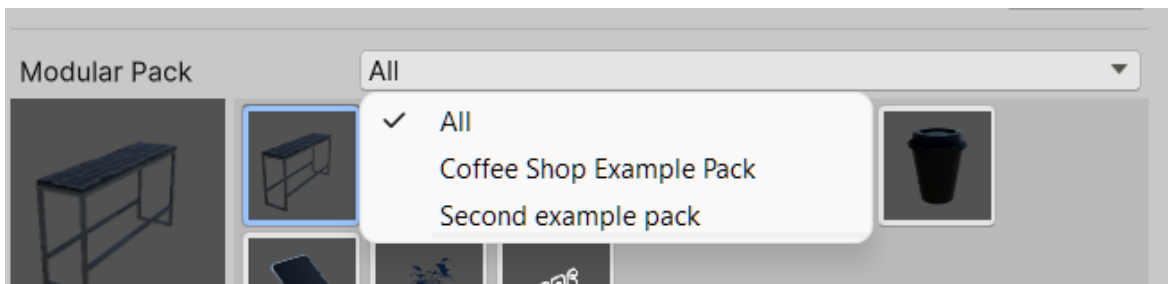
4. **Decorator Tool:** LMB (Accept Action) will not proceed if the cursor is over the prefab's gizmos.



Render Pipeline Compatibility

MBS is the level design tool, the only purpose of which is to build structures out of users' prefabs. MBS doesn't touch the rendering process in any way. **MBS is compatible with any Unity Render Pipeline.**

MBS v0.2 includes 2 modular packs for example purposes: Coffee Shop Example Pack and Second example pack. They are added to show the capabilities of MBS, how flexible it is, and how it can be used. Both modular packs contain different prefabs, textures, and standard materials. **To make it work with any other Render Pipeline please use [Unity's built-in material upgrading tool](#).**



From Developer

- Thank you for purchasing the MBS, and thus for supporting me, and my project.
- If you like MBS, I would be grateful if you take the time to leave a review in the asset store!
- I'm not a native English speaker, and to write docs and other info about MBS I use my own experience in English and Google Translate app, so, if you found some errors or unusual, incorrect use of language or something that is hard to understand - please inform me about this via email.

Roman Ponomarenko,

Thank you